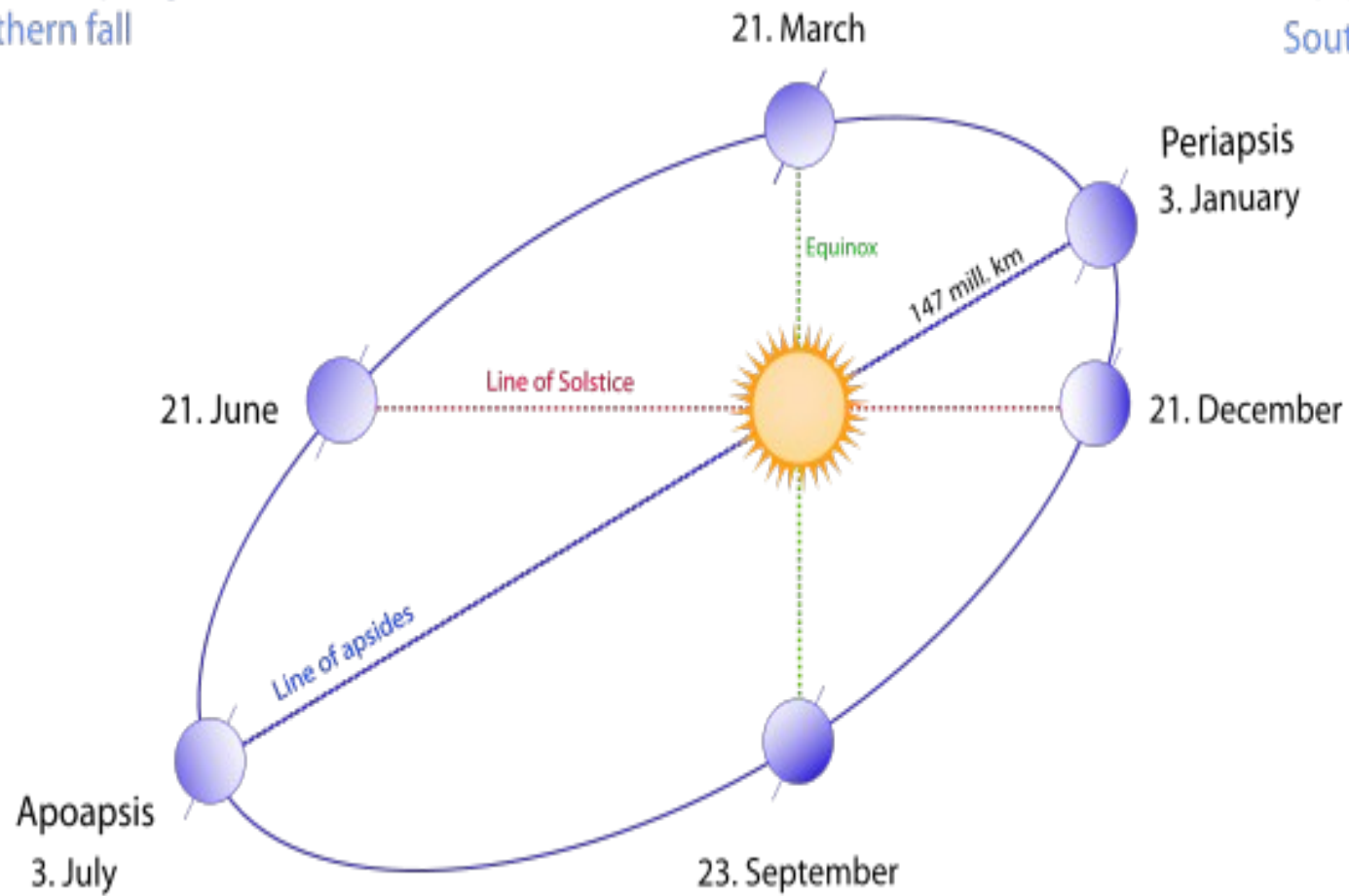


Review and Celestial sphere.



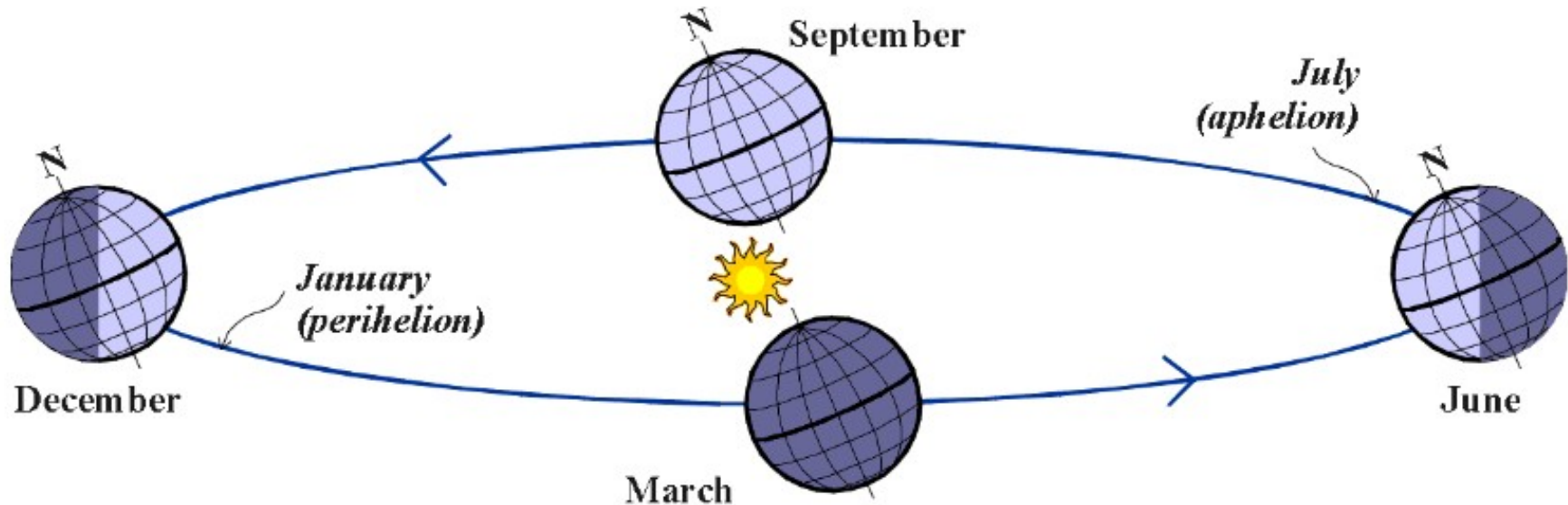
Northern spring/
Southern fall

Northern winter/
Southern summer



Northern summer/
Southern winter

Northern fall/
Southern spring



The Earth and the other planets, all, are part of the ecliptic plane. The plane is part of the Milky way

→ show picture ecliptic_plane.png

→ show milky way.

→ show ecliptic_plane_constellation All the planets and the zodiac constellations Are in the ecliptic plane. We see them on an imaginary line. Ecliptic line

→ show zodiac_constellation_belt

→ See how the Earth is tilted. Show ecliptic_3d.gif and precession-North-Pole-circle

→ Other planets are tilted too. Show spins_planets

→ show celestial_sphere3 . The 2 planes. Equatorial and ecliptic as viewed from Earth.

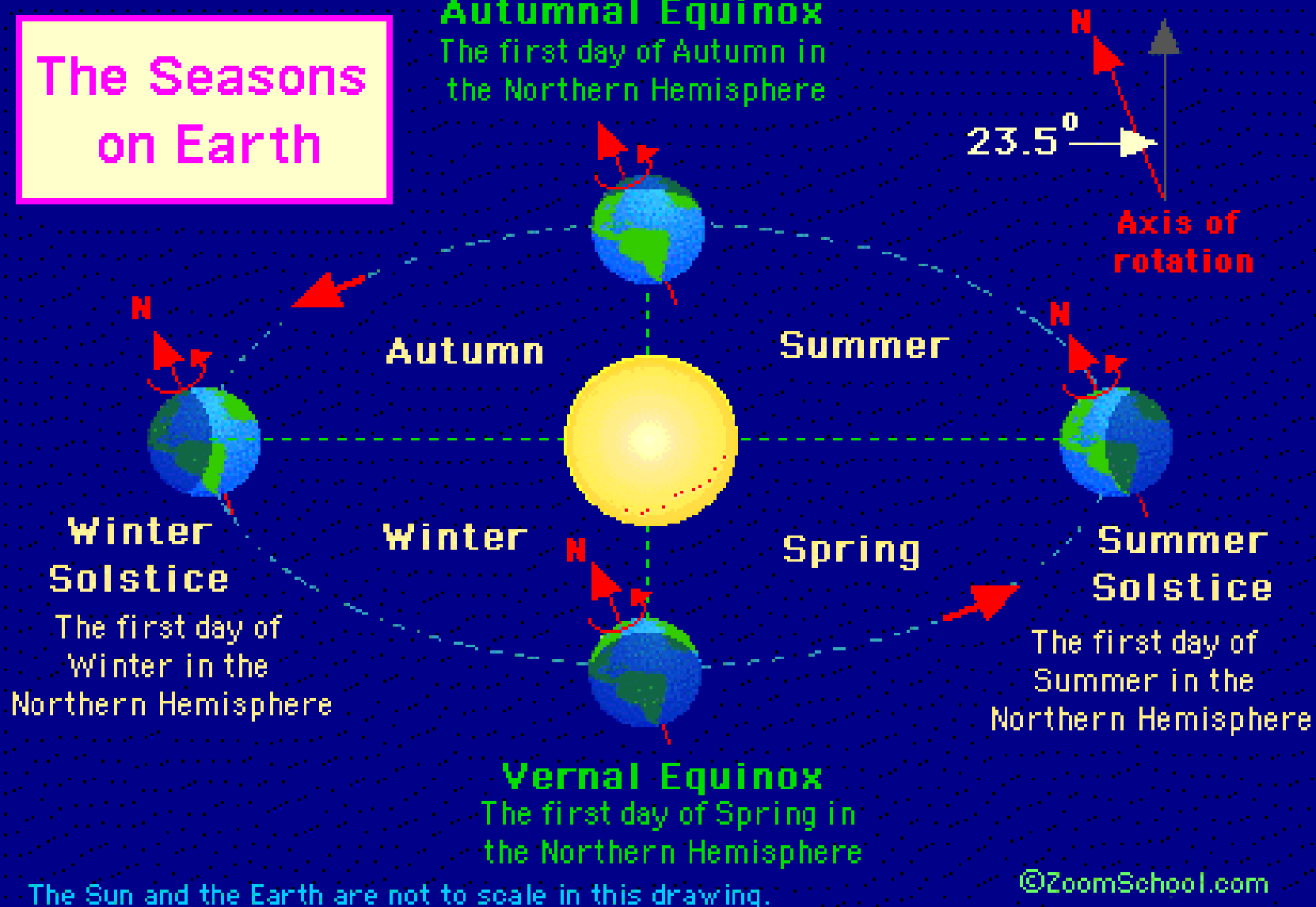
You see in July the Earth is toward the Sun → longest day. (North hemisphere)

... comment.

The Seasons on Earth

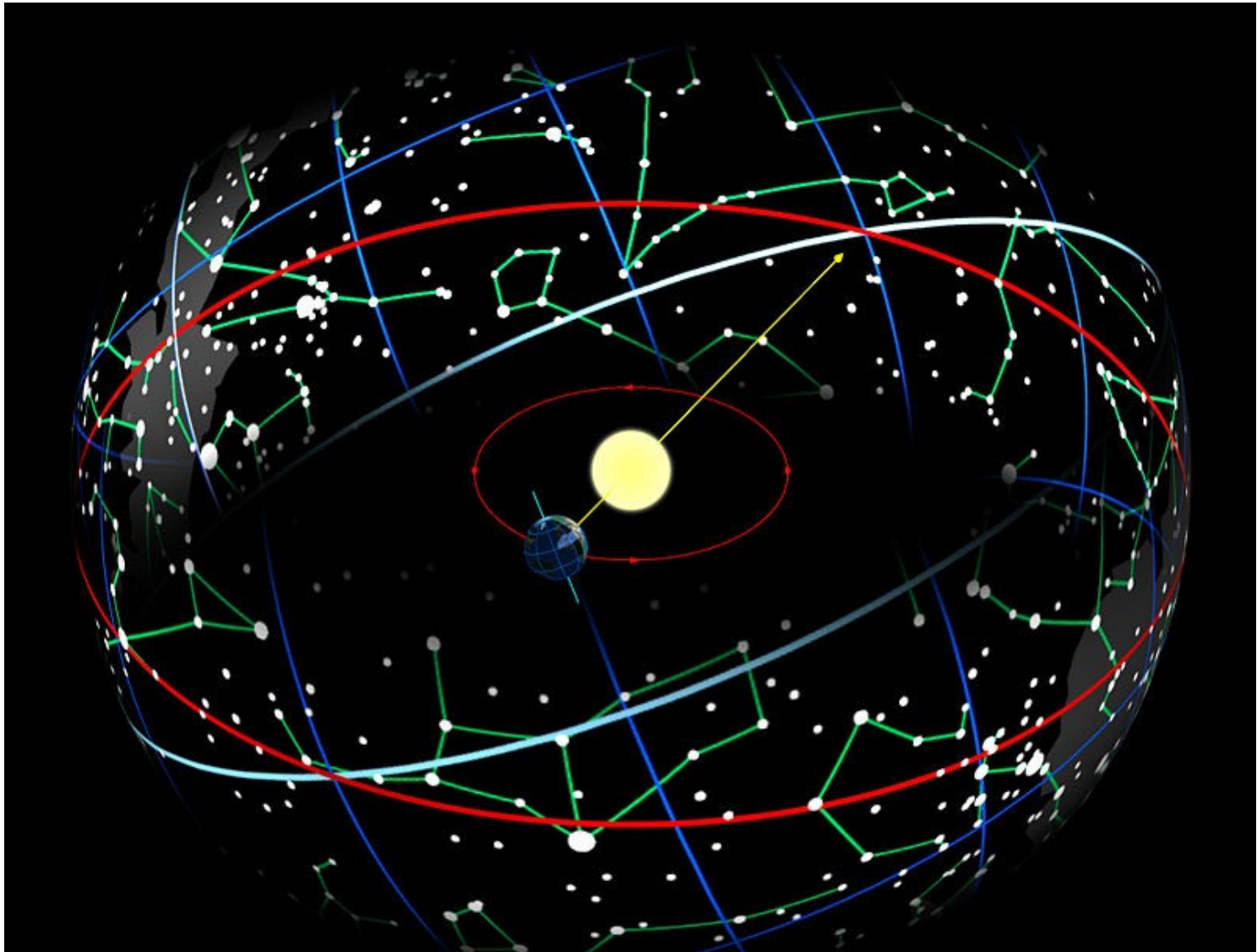
Autumnal Equinox

The first day of Autumn in the Northern Hemisphere

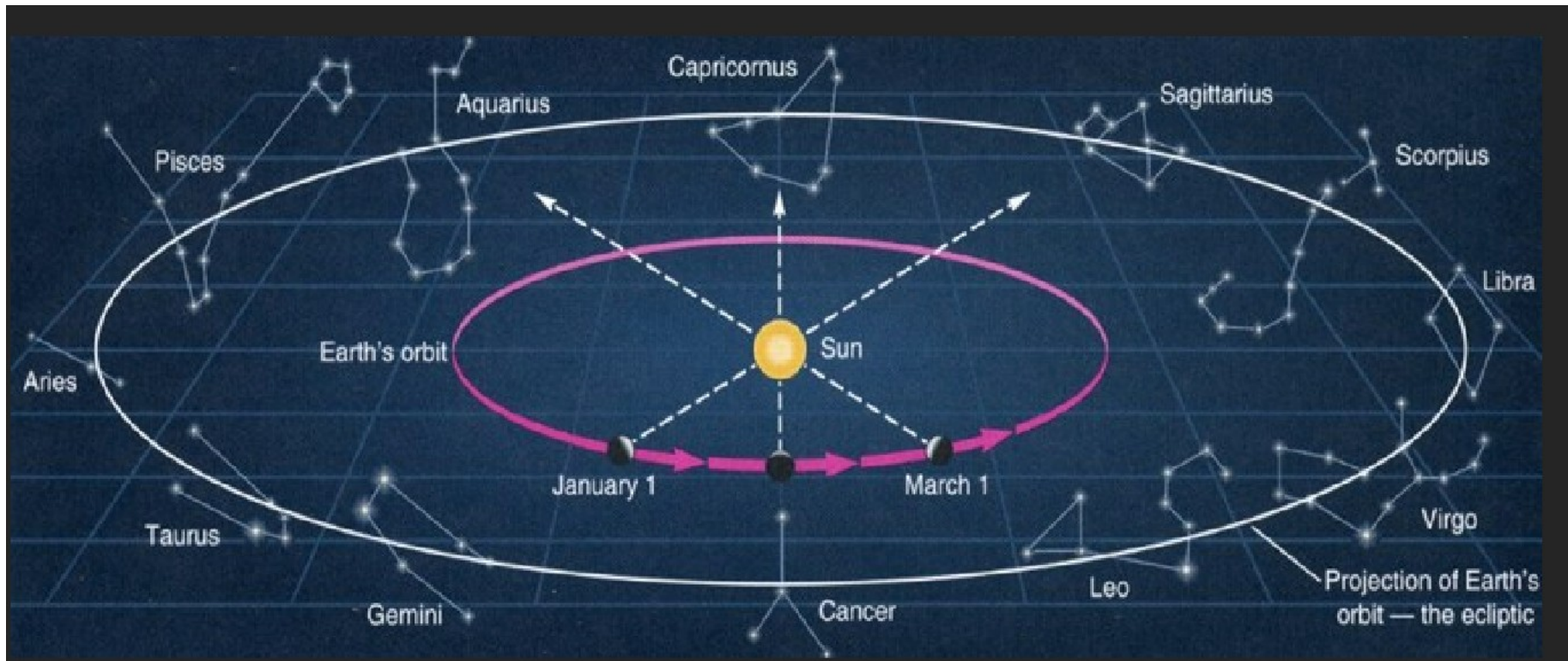


March 20 is the Vernal equinox. (around there).
Vernal equinox is special. See next slide.

The ecliptic is in red and as a dashed line on your planet locator. The zodiac is a zone (18 degrees) centered on the ecliptic. The equator line is in blue. They don't match because the earth is tilted. The ecliptic plane includes the sun and the planets.



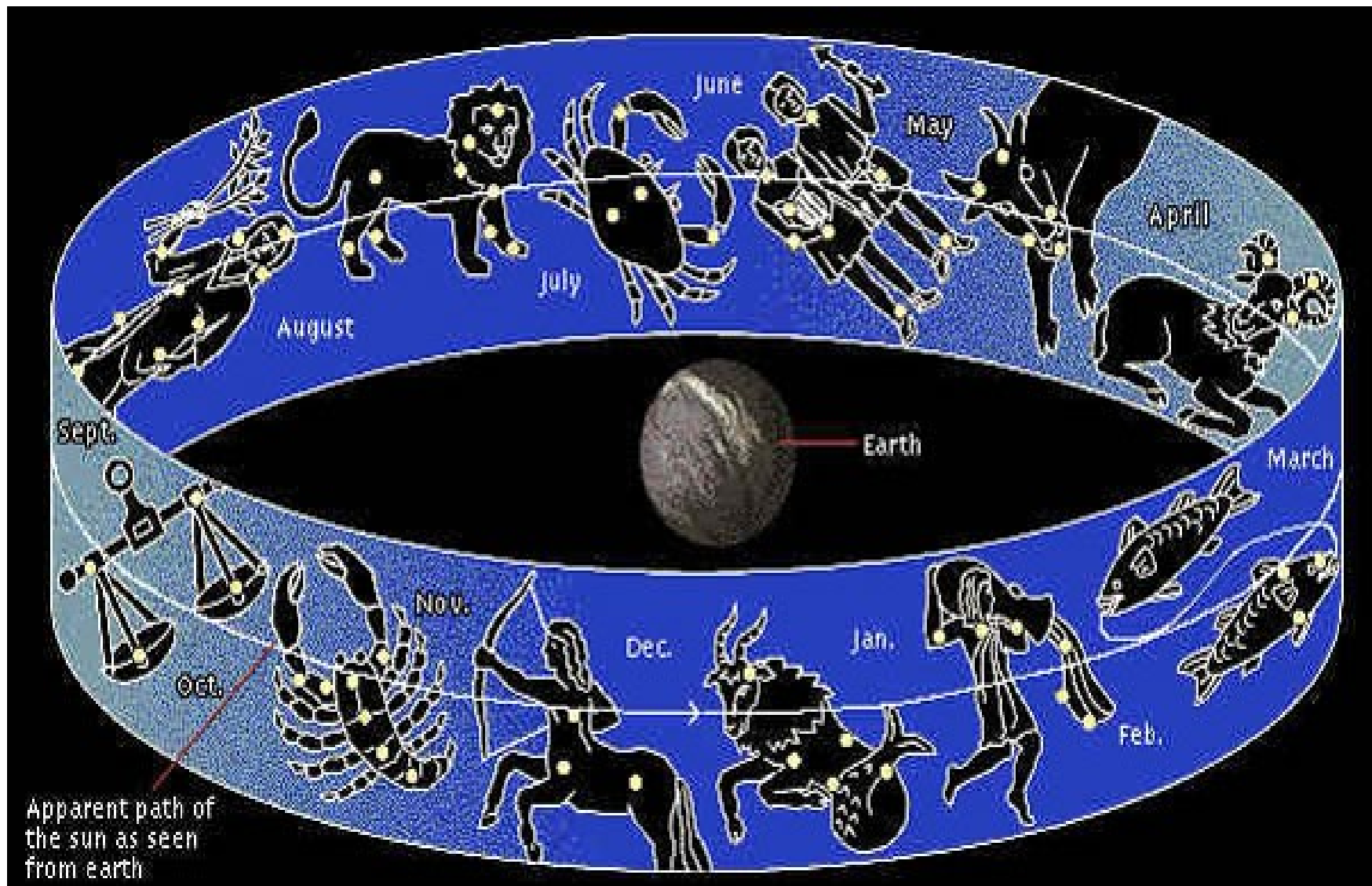
See slide 11



The zodiac constellations are along the ecliptic plane. Your sign is determined by projection on the sun on that line. March 20 is the vernal equinox. It's the sign Pisces. But because the tilt of the earth is changing. It will point to aquarius. Therefore the famous song "age of aquarius" from musical Hair. "Age of love, light, and humanity " (lol)

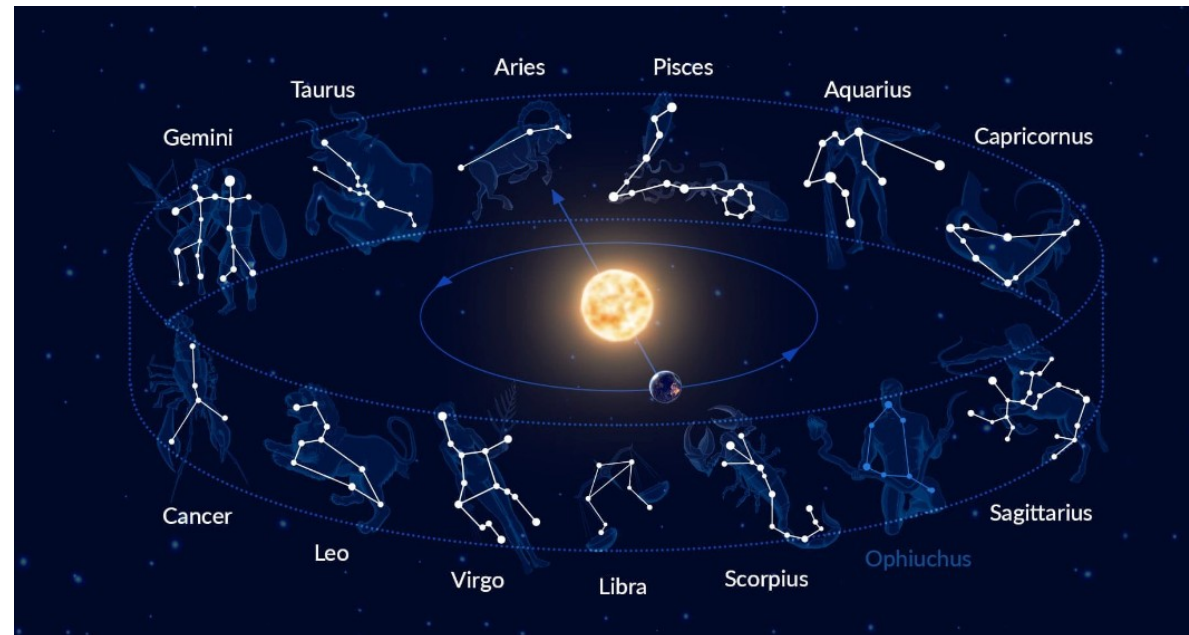
https://www.youtube.com/watch?v=ajgeaOt_HTQ

Show picture



Show picture zodiac2

Ophiuchus the 13th zodiac sign ? The forgotten one? Between Sagittarius and Scorpius Nov 29 and Dec 19

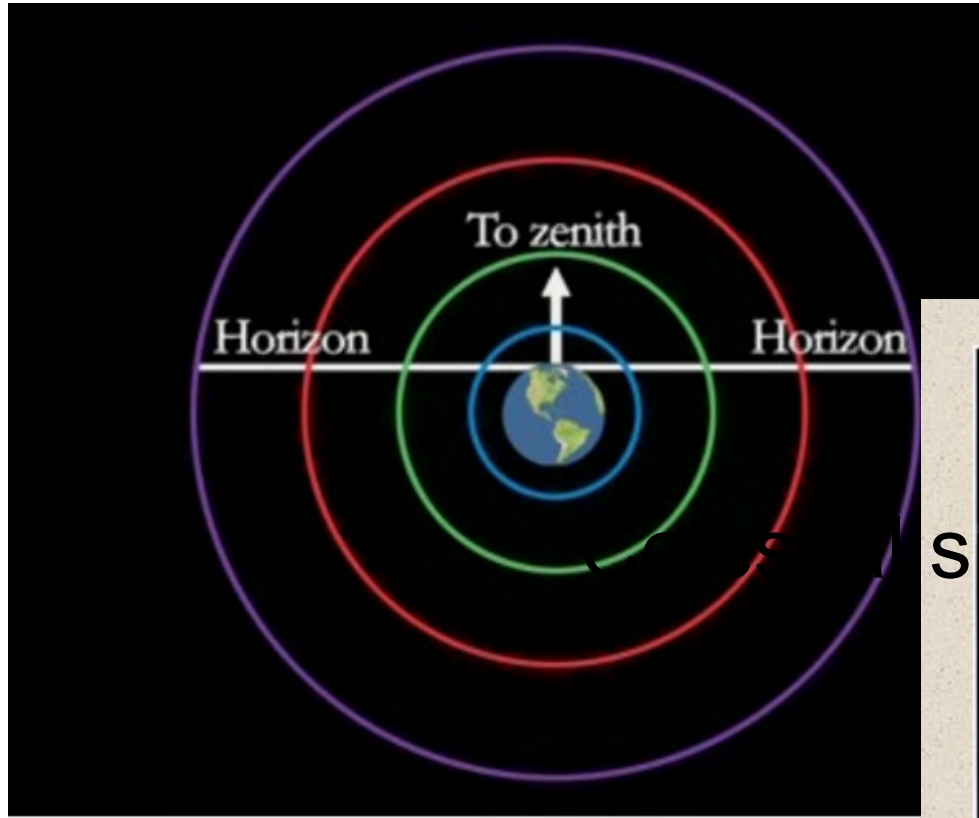


<https://www.almanac.com/what-age-aquarius>

In ancient Greece, Ophiuchus represented Asclepius, the god of medicine and the son of the god Apollo. In one version of this story, **Asclepius killed a snake with his staff. But another snake dropped some herbs on the dead one, bringing it back to life.** Jun 4, 2018

<https://stardate.org/radio/program/2018-06-04#:~:text=To%20appease%20him%2C%20Zeus%20placed,of%20the%20story%20of%20Ophiuchus.>

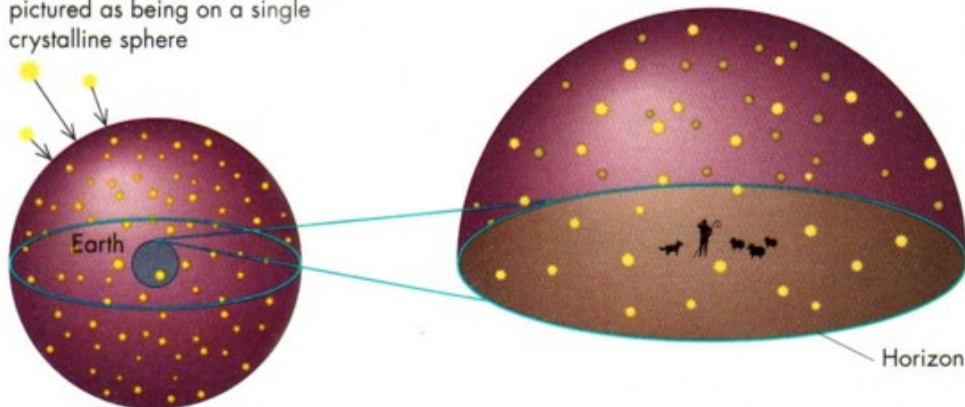
We see half a circle if the stars are infinitely far.



Celestial sphere.

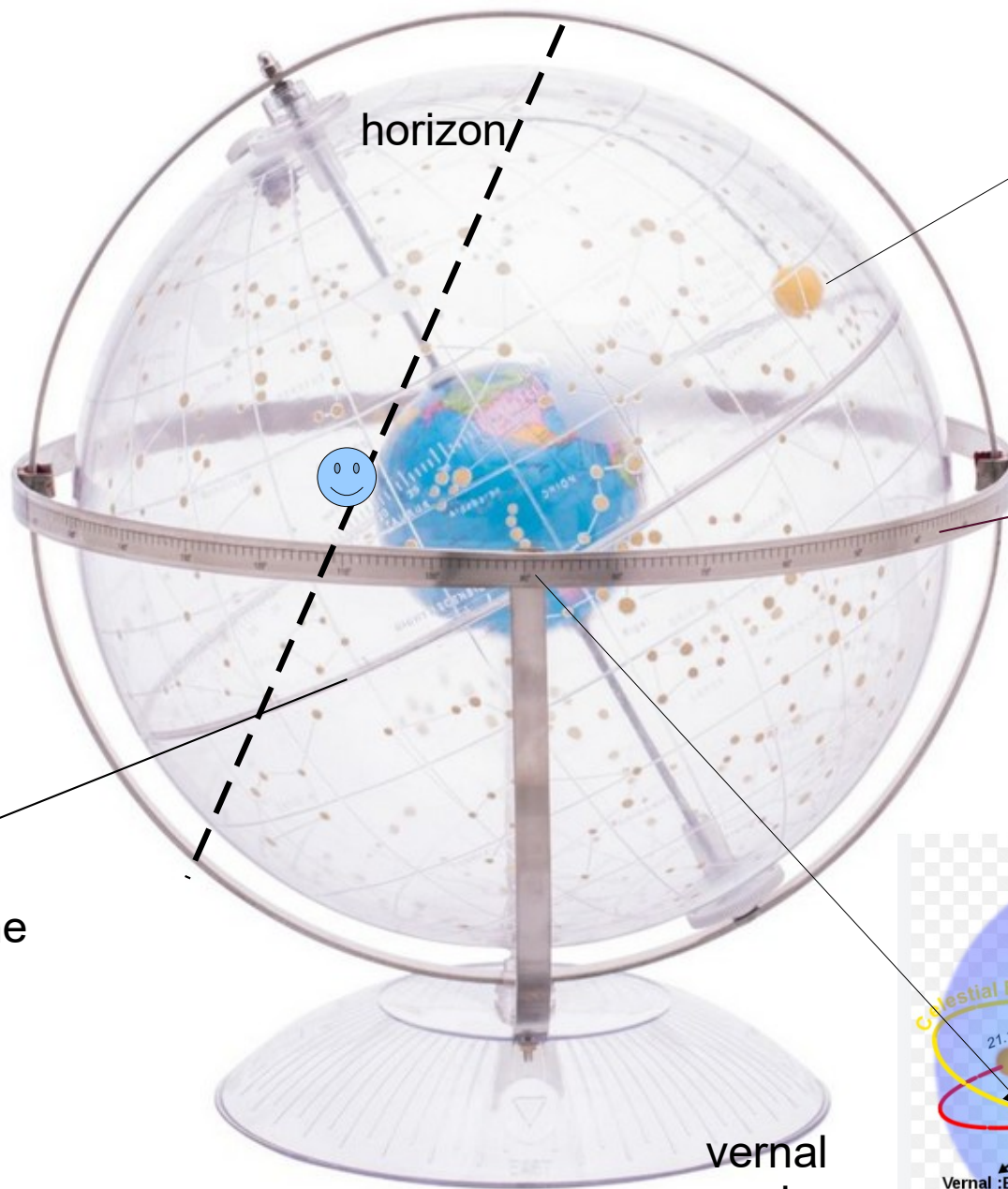


Stars, no matter how distant, are pictured as being on a single crystalline sphere



Model: The celestial sphere

The human experience of the celestial sphere



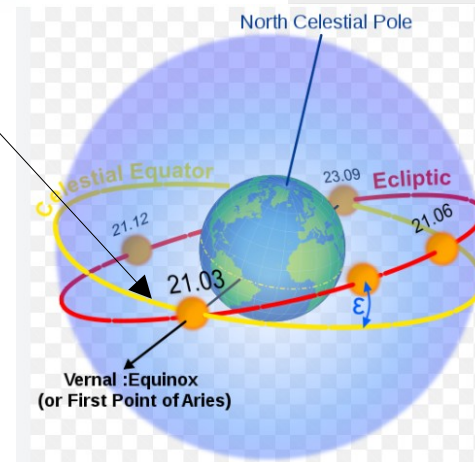
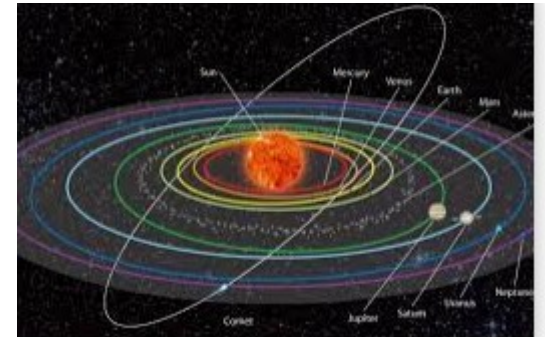
horizon

Sun

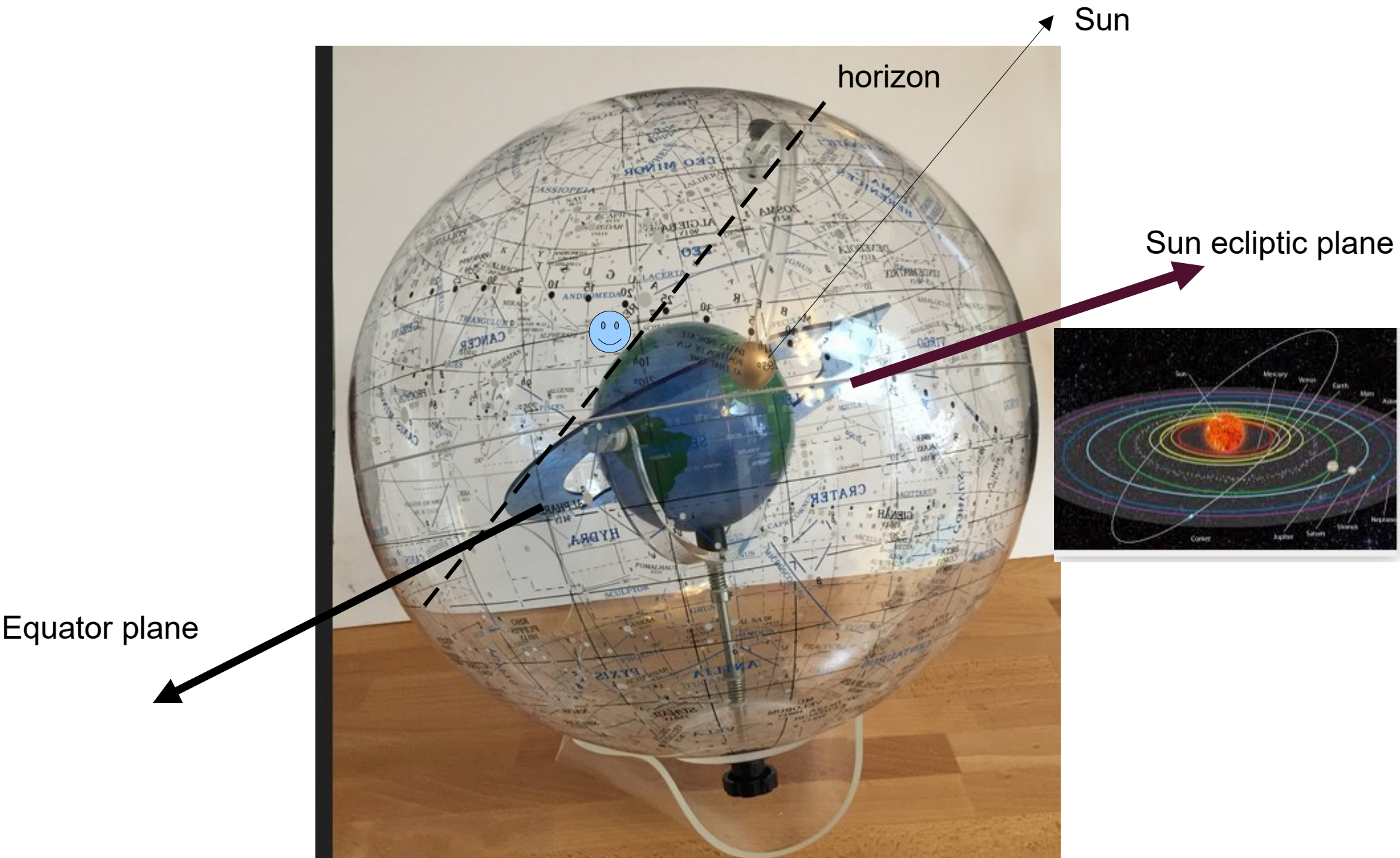
Ecliptic line →
Ecliptic plane

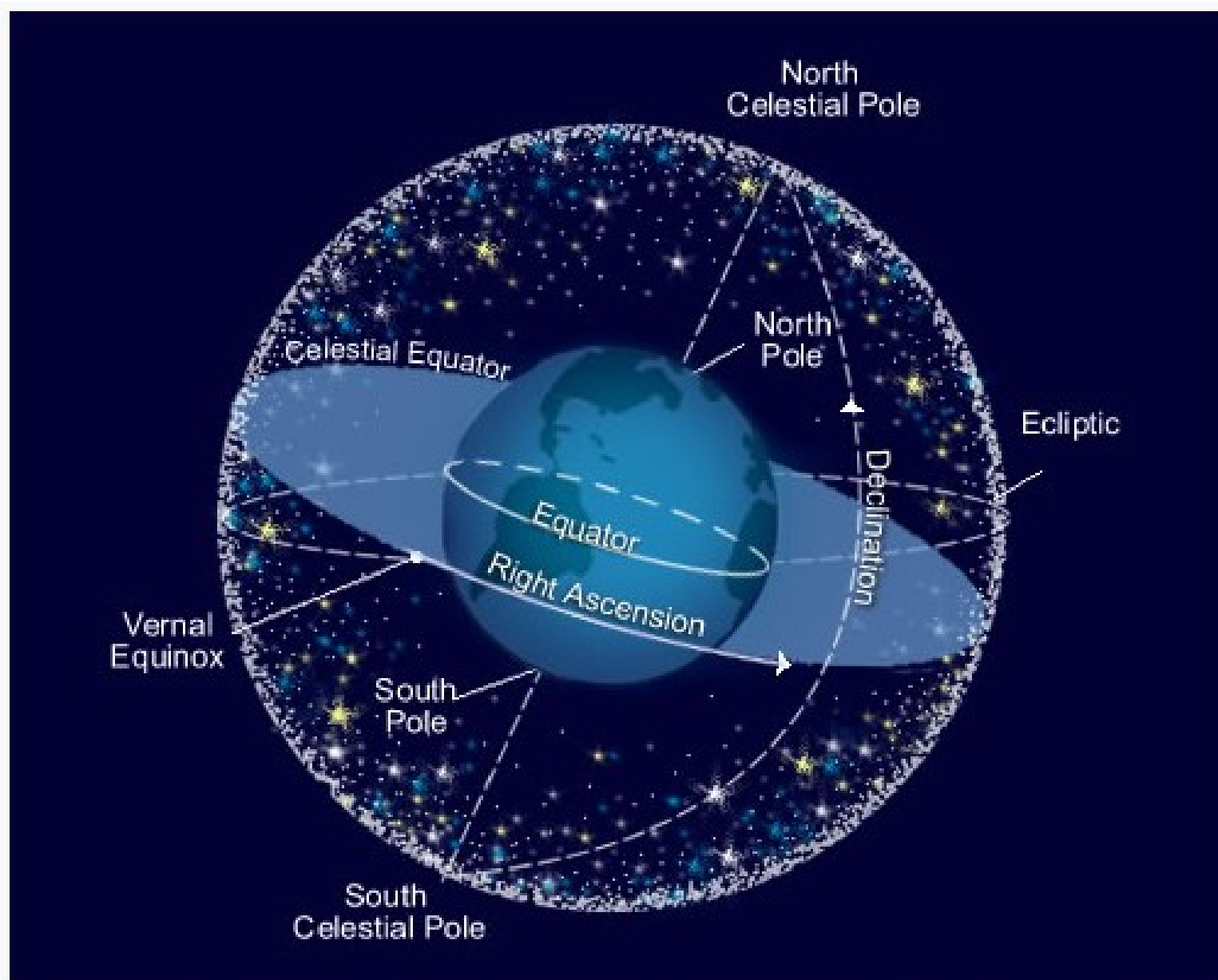
Equator plane

vernal
equinox

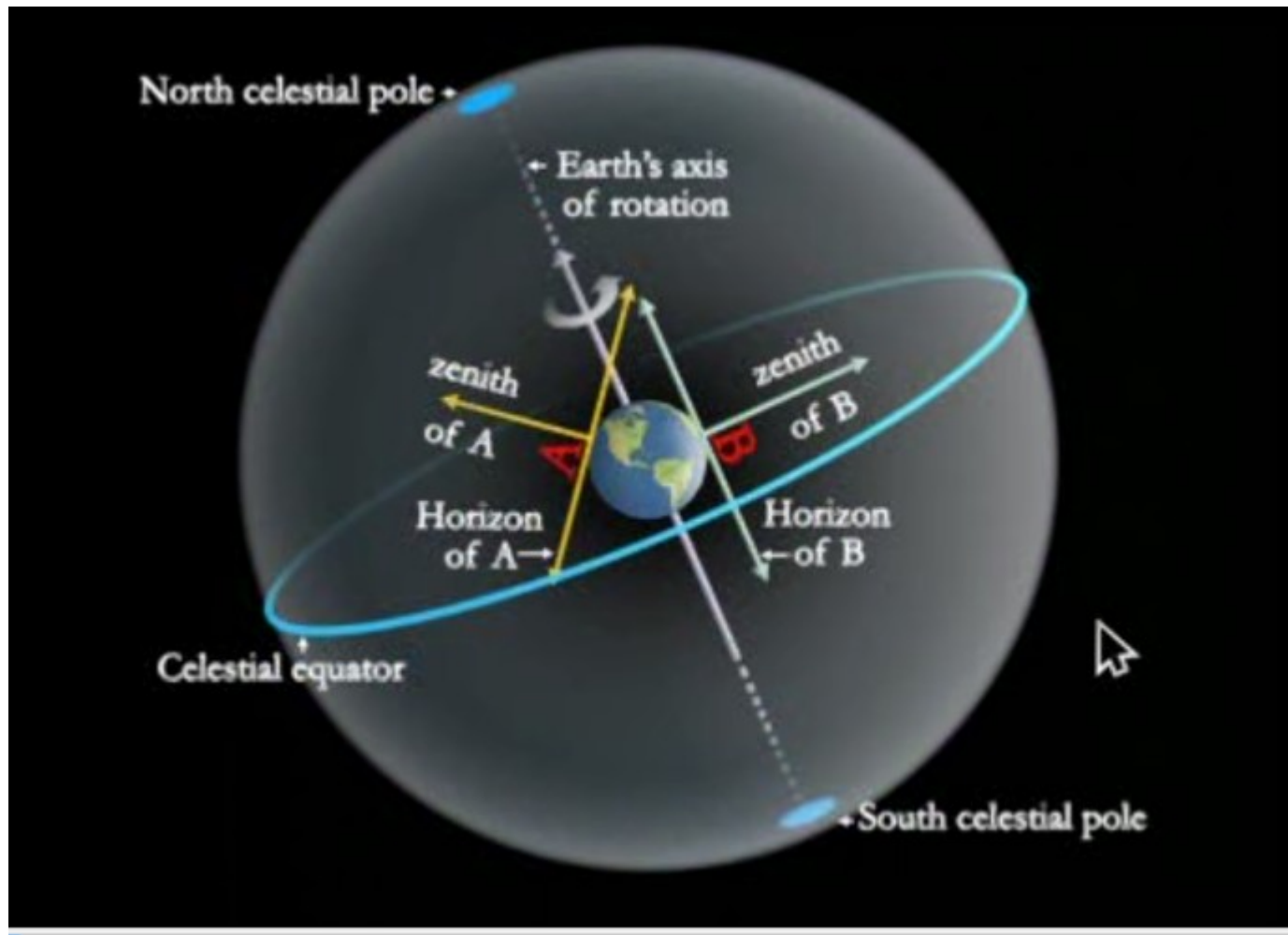


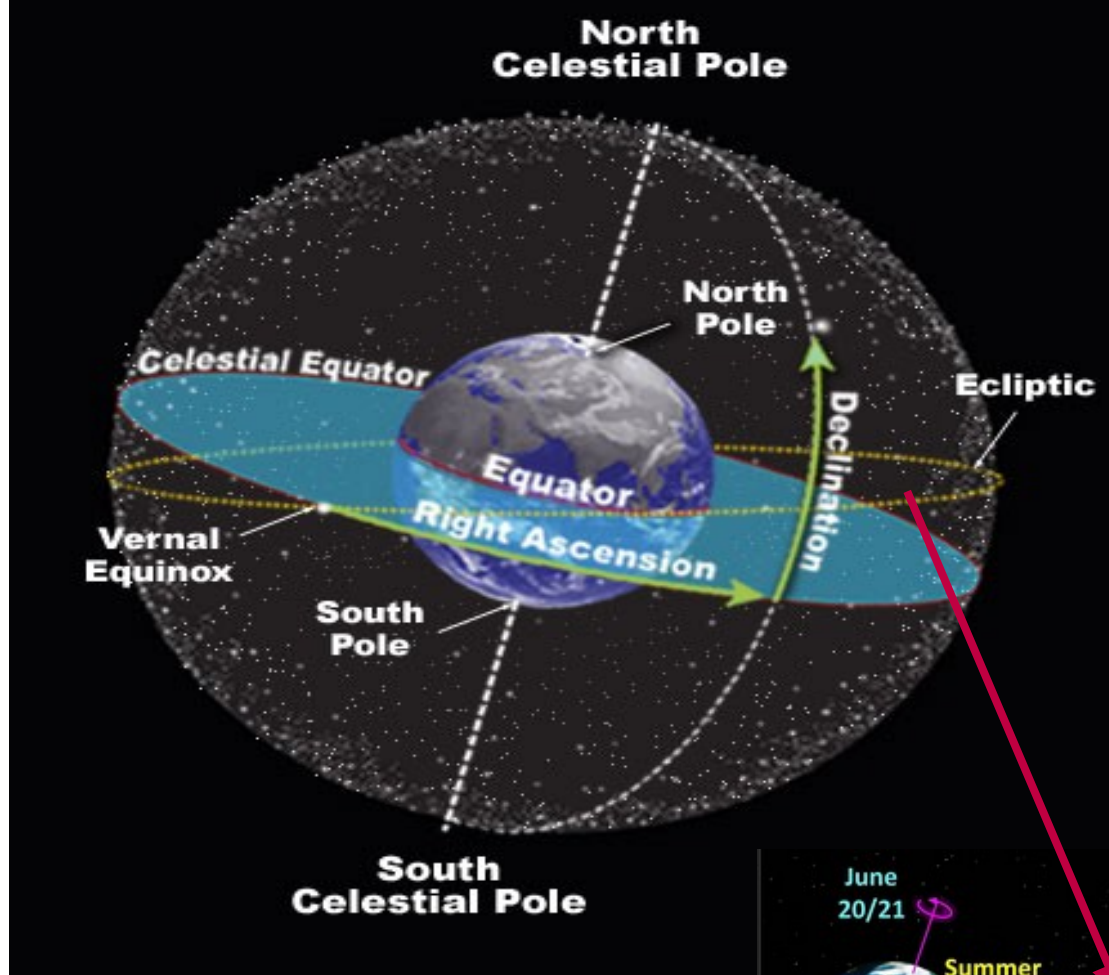
<https://sciencesims.com/sims/celestial-sphere/>



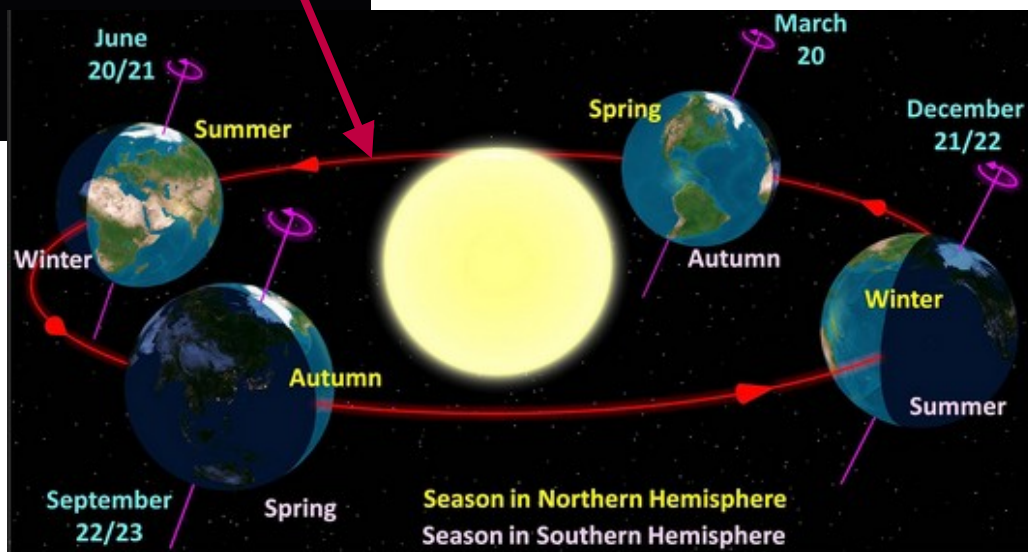


See how,, if you are at the equator, polaris is just above the horizon!
At the North pole, polaris is above your head.
Elsewhere, it depends on the latitude . See video?



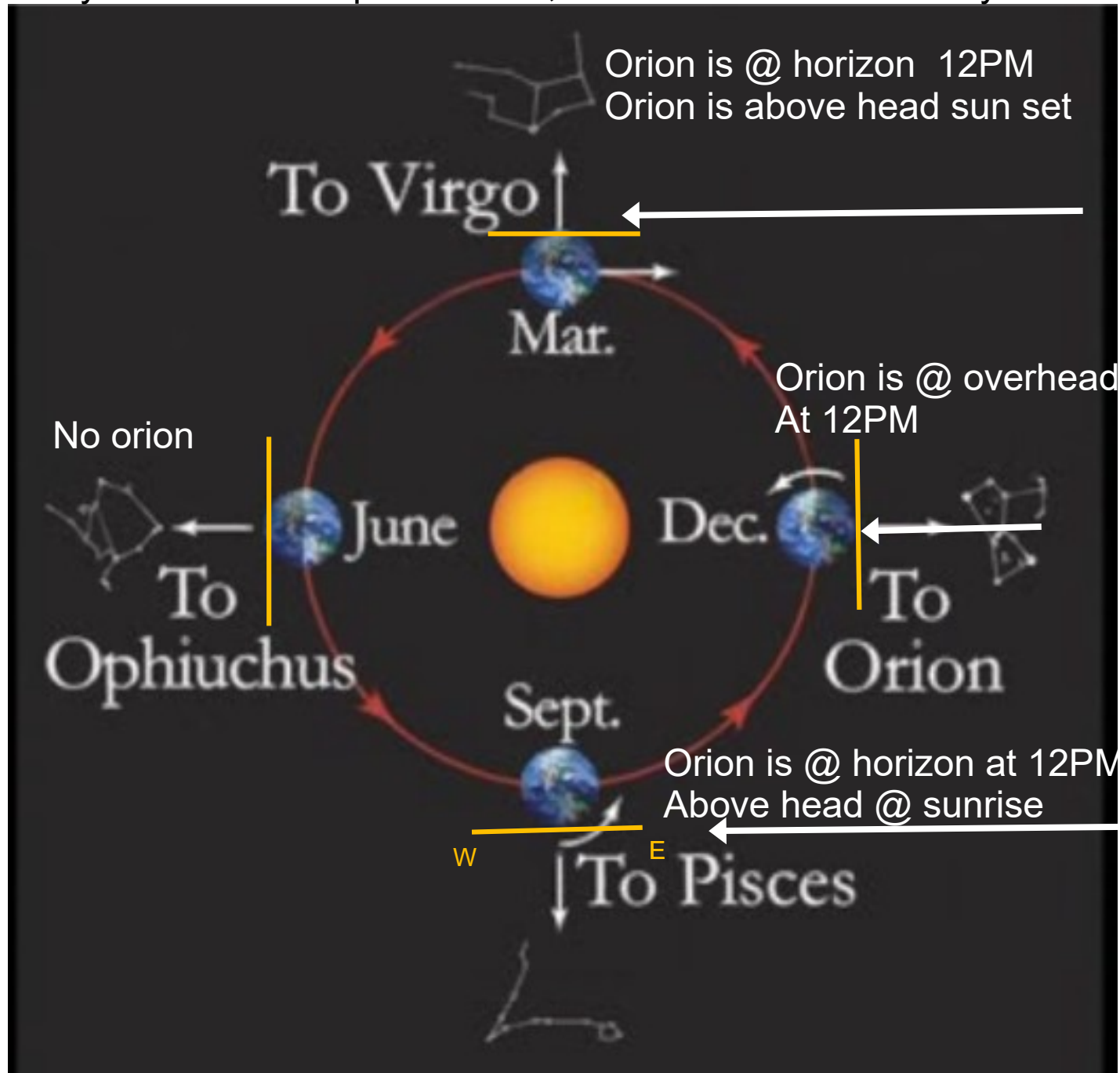


WATCH 2 VIDEOS. The history one, (1:25 skip)

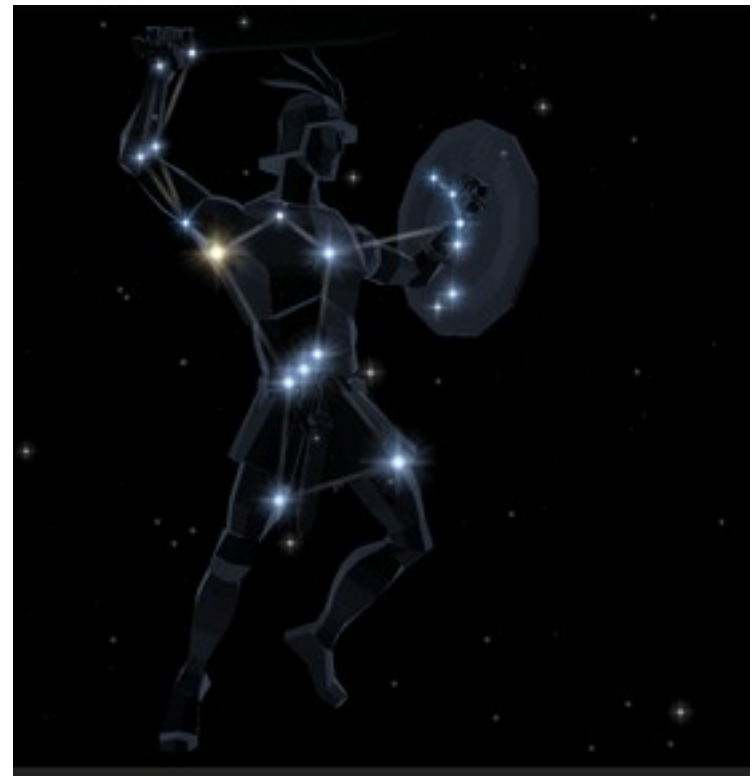


<https://www.timeanddate.com/astronomy/night/usa/miami>

Suppose you are at the equator. June, orion is not visible. Its day time. Can't see it.

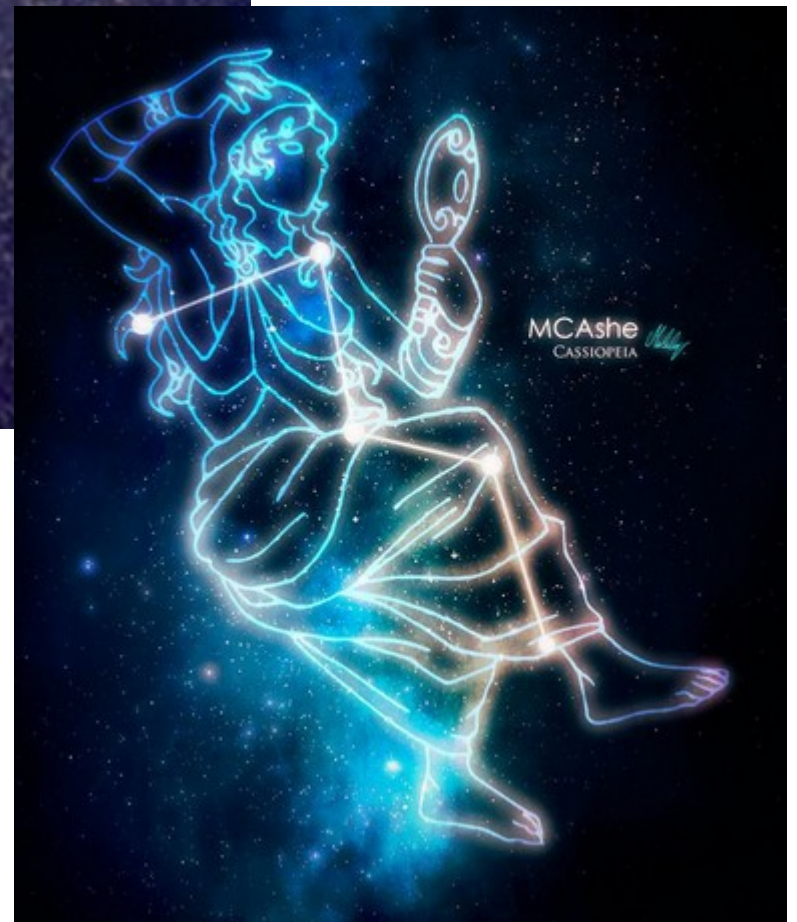


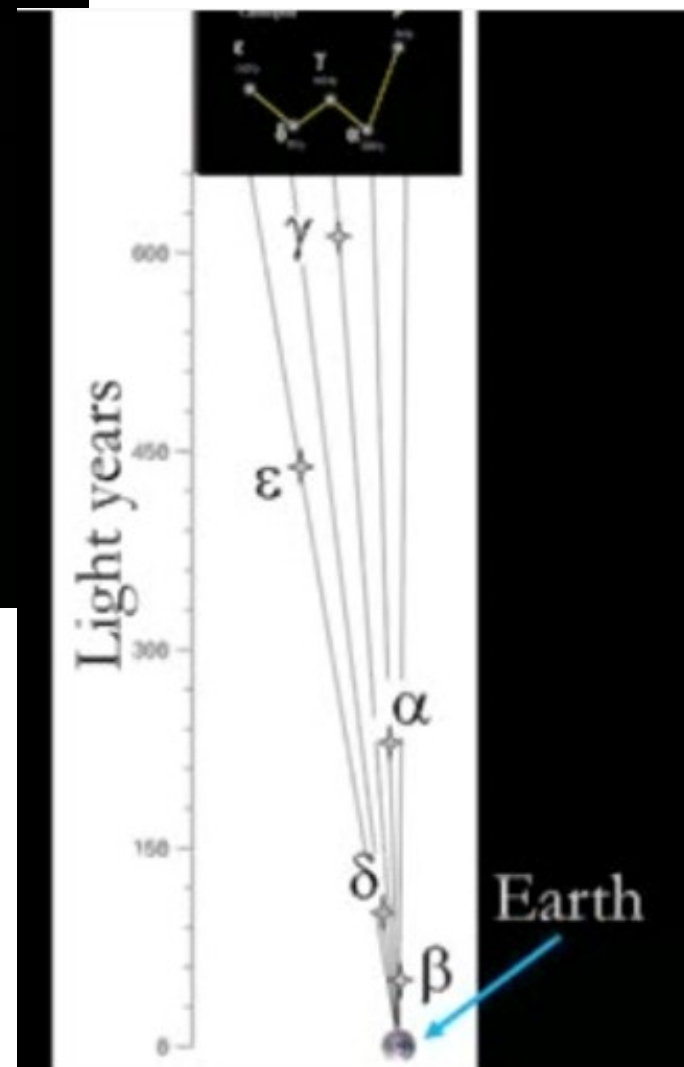
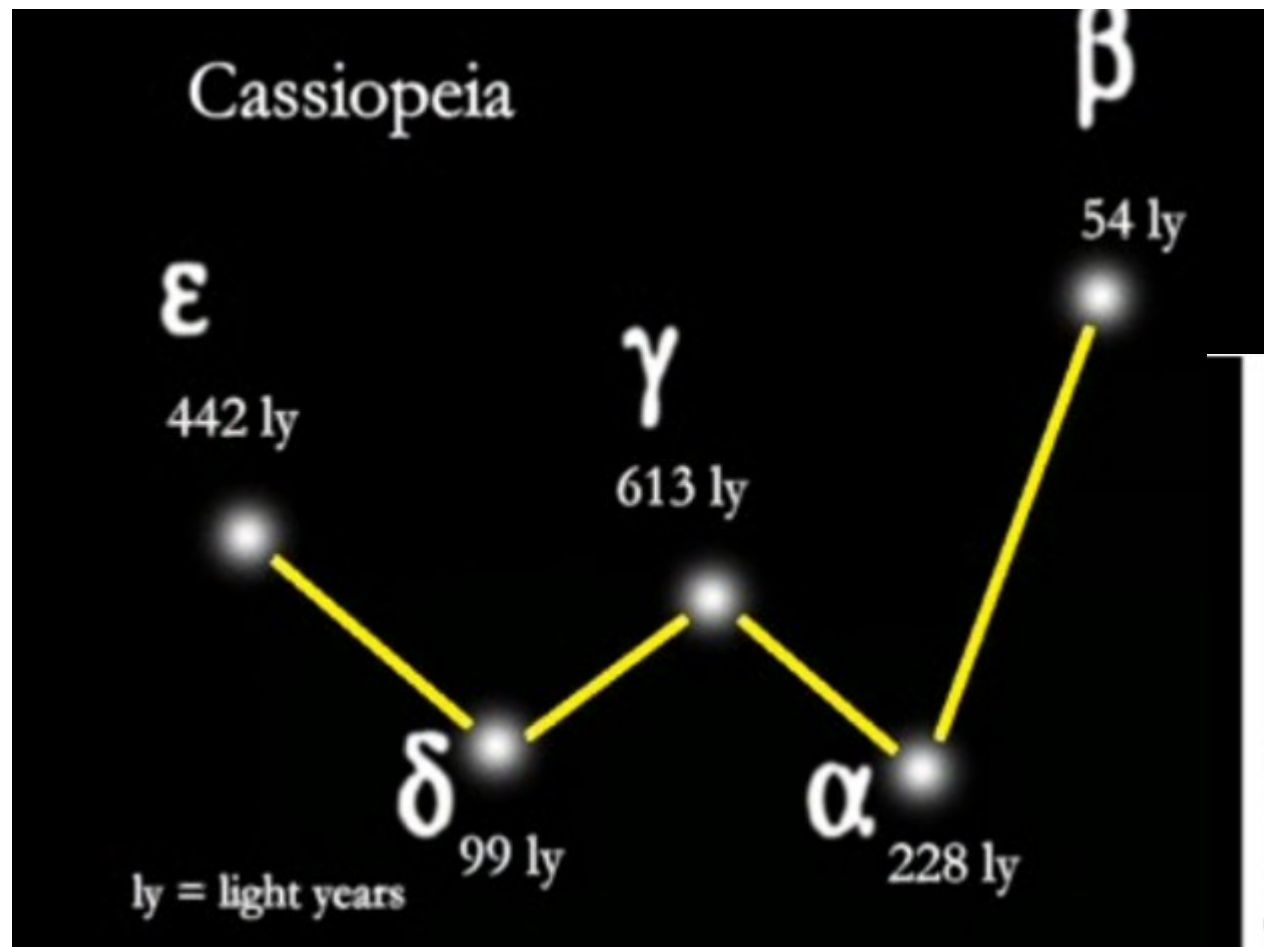
Videos: Why do stars move, winter hexagon,

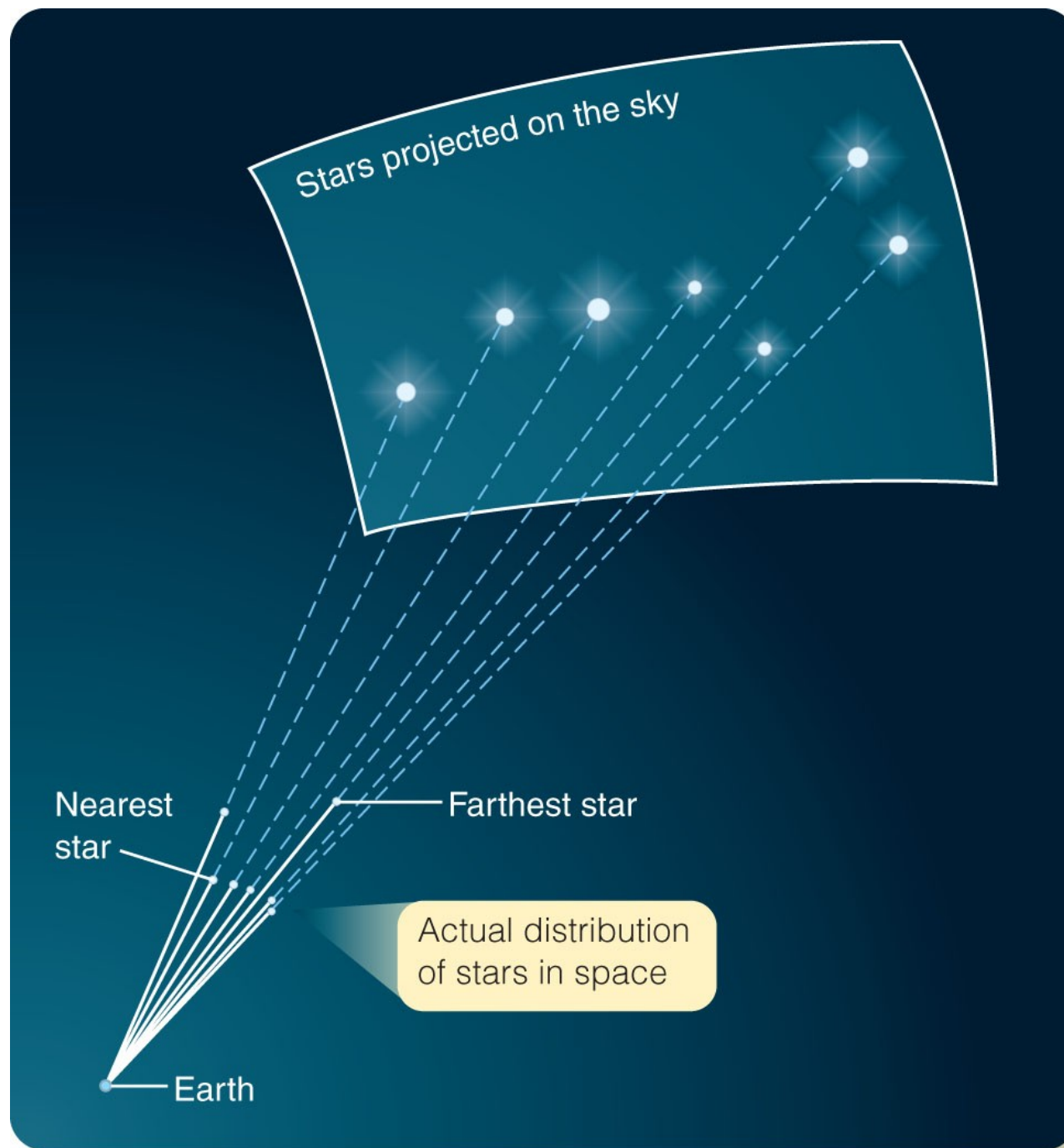


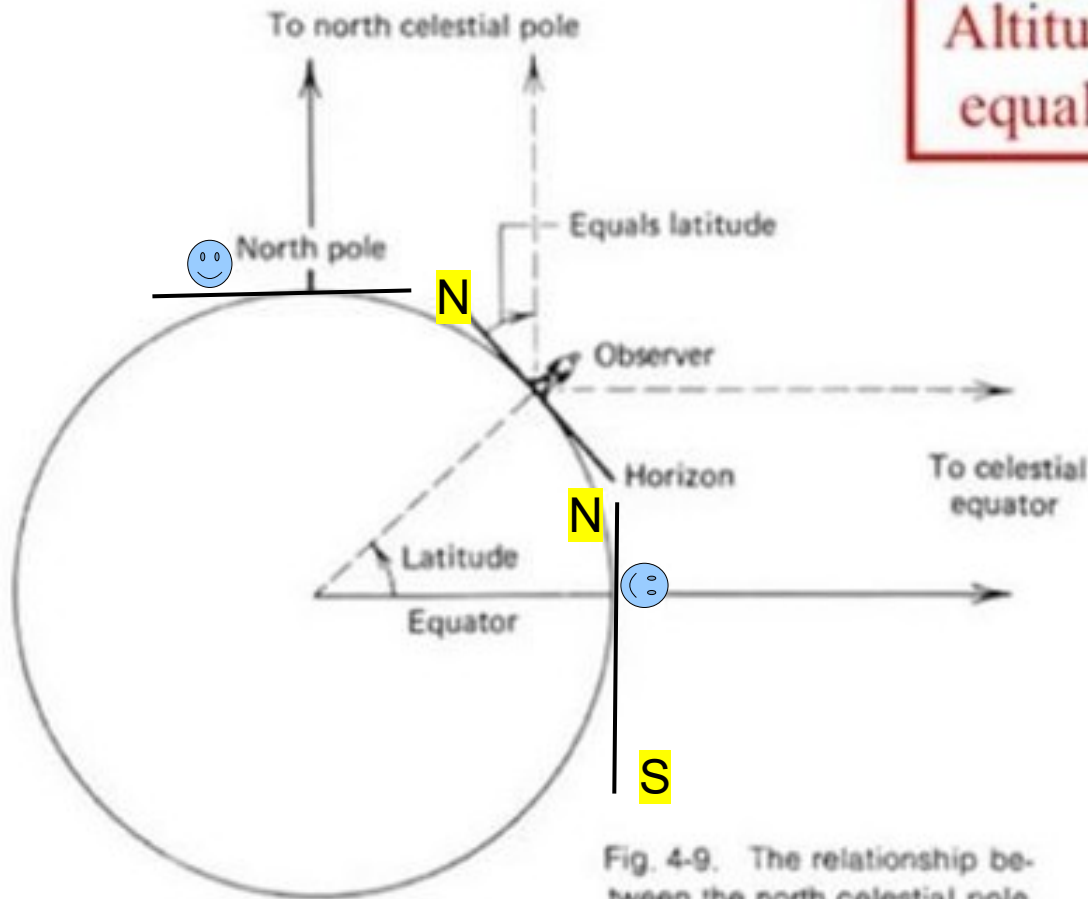








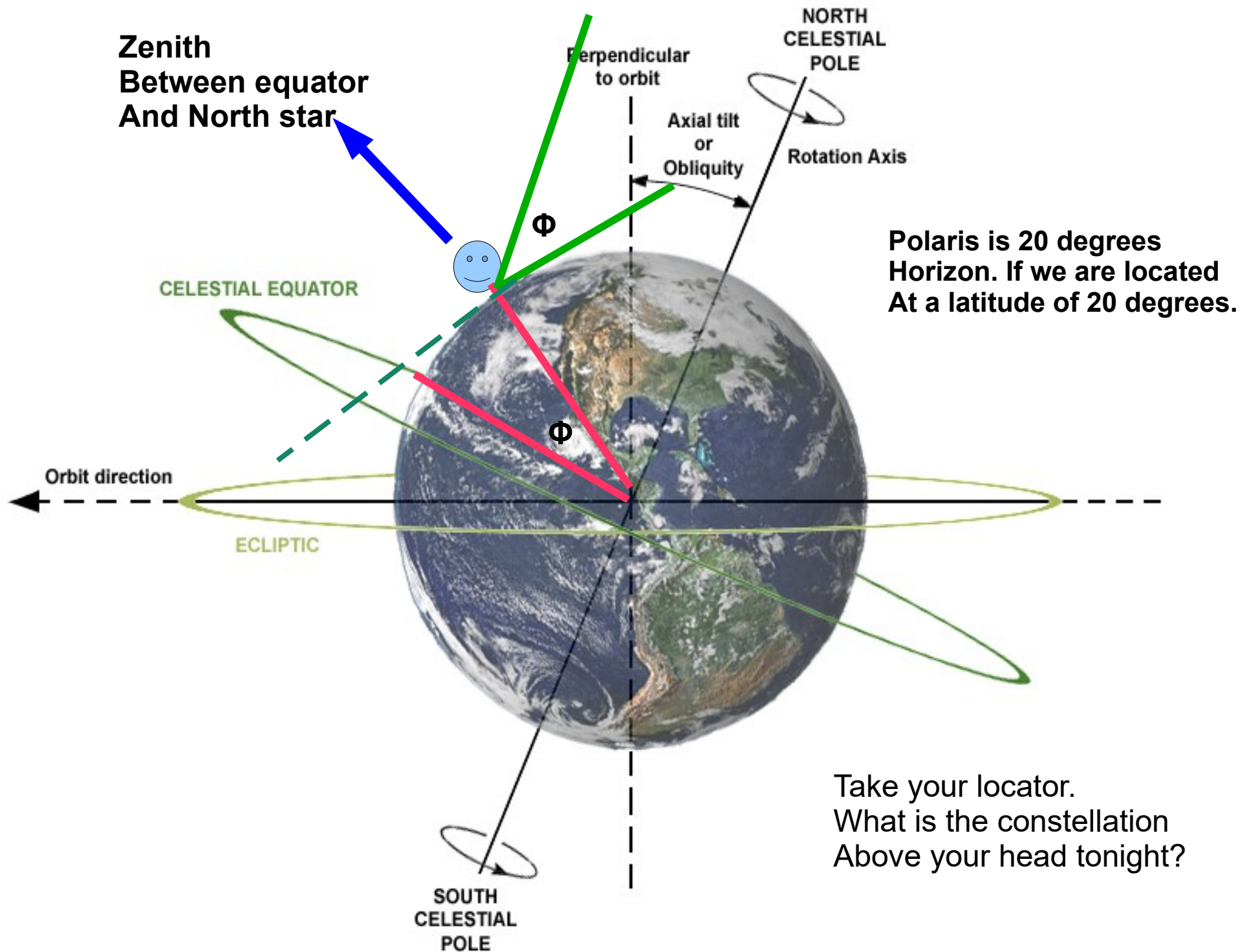




Altitude of the north star
equals our latitude

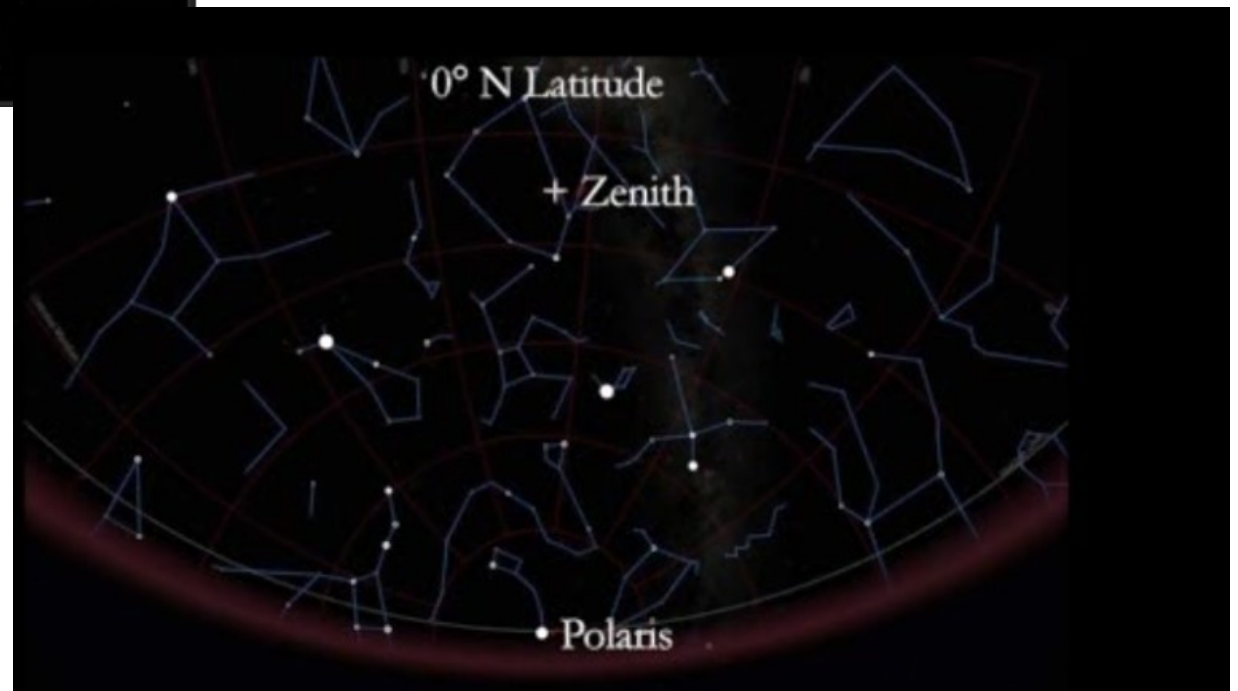
Celestial equator
maximum altitude
and angle at horizon
is 90° -latitude

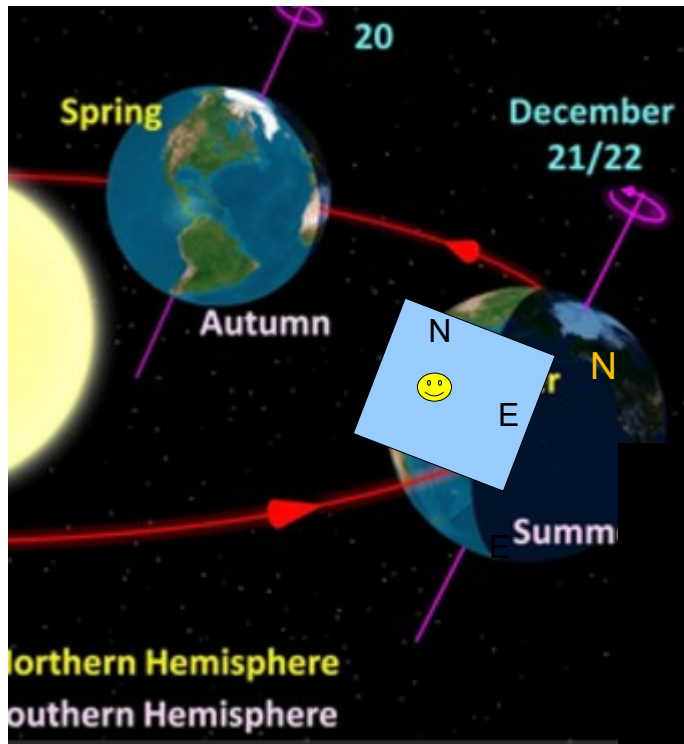
Fig. 4-9. The relationship between the north celestial pole, the terrestrial and the celestial equator, and the latitude of an observer.



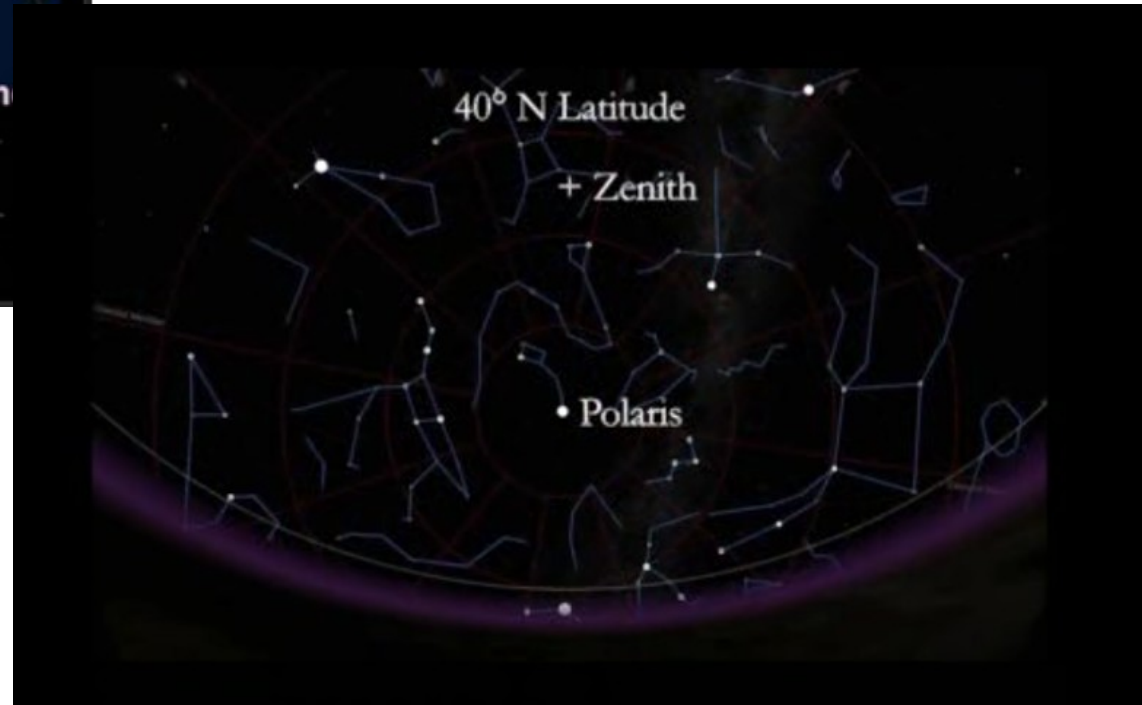


At the equator polaris is seen in the horizon Plane. (parallel rays coming from stars).





Above the equator
 Polaris is seen above the horizon
 Plane. (parallel rays coming from stars).



(Miami
 25)

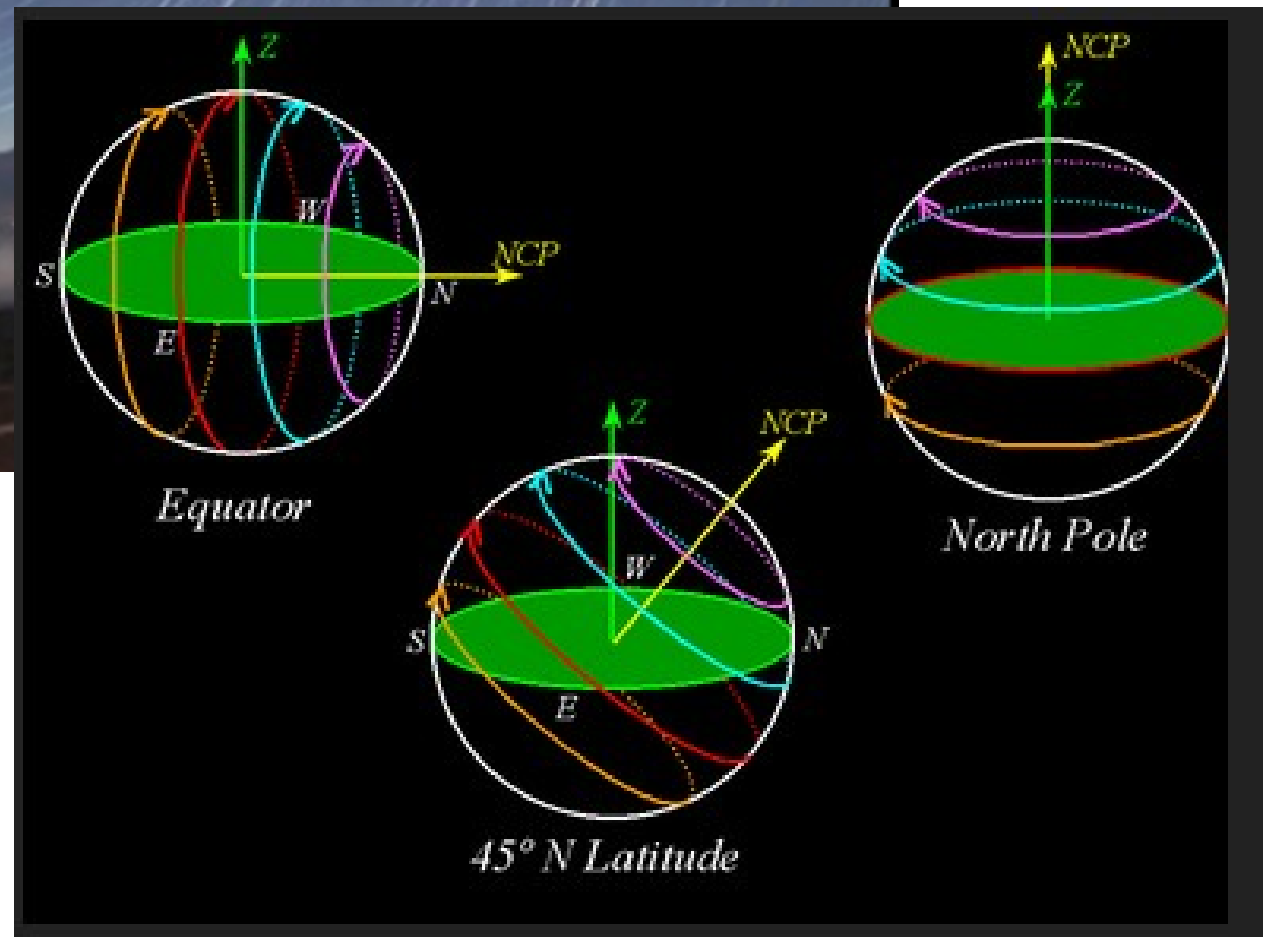


At the pole polaris is seen above head
(parallel rays coming from stars).

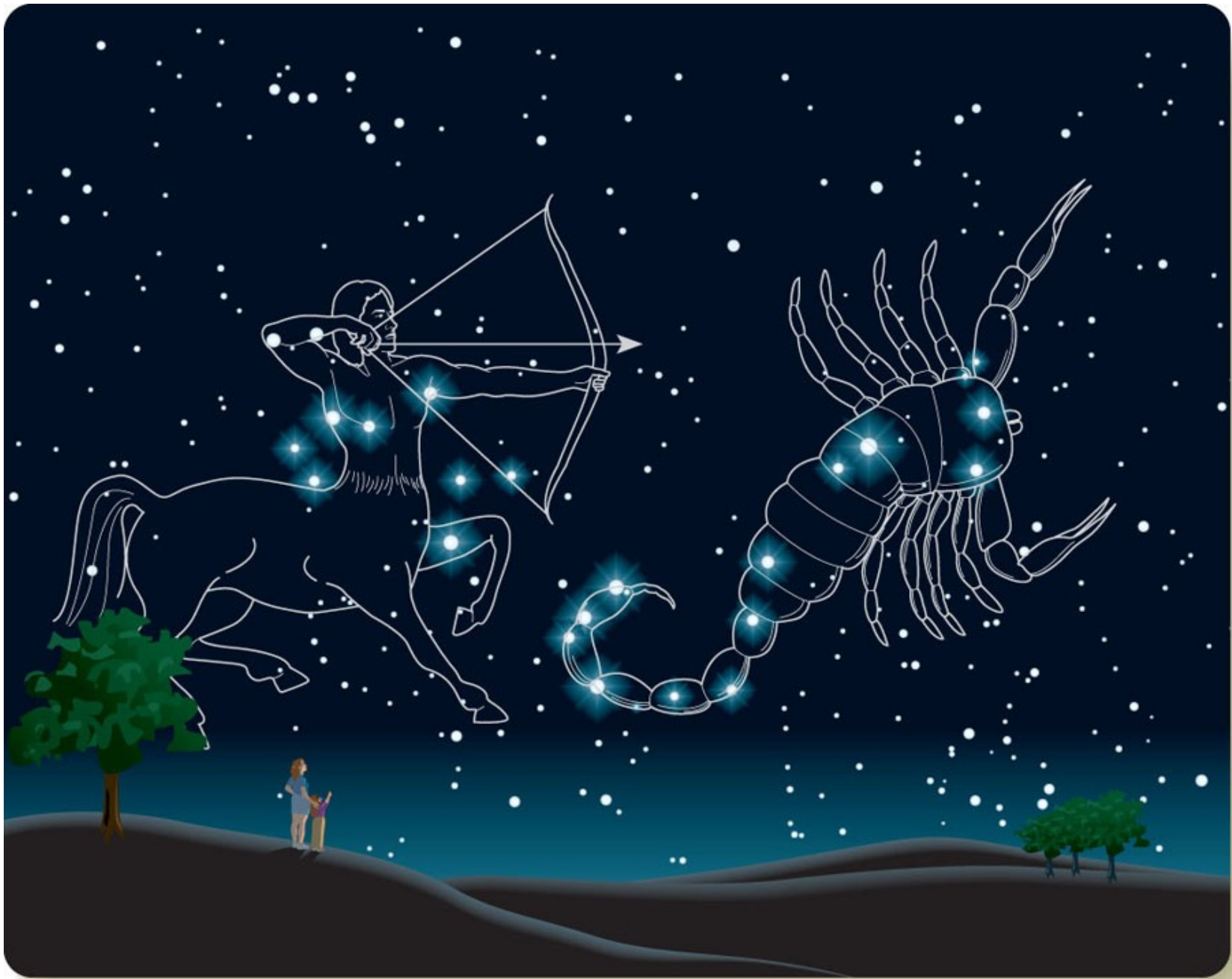
Show polaris in Miami sky
Above the horizon
(20 degrees about)



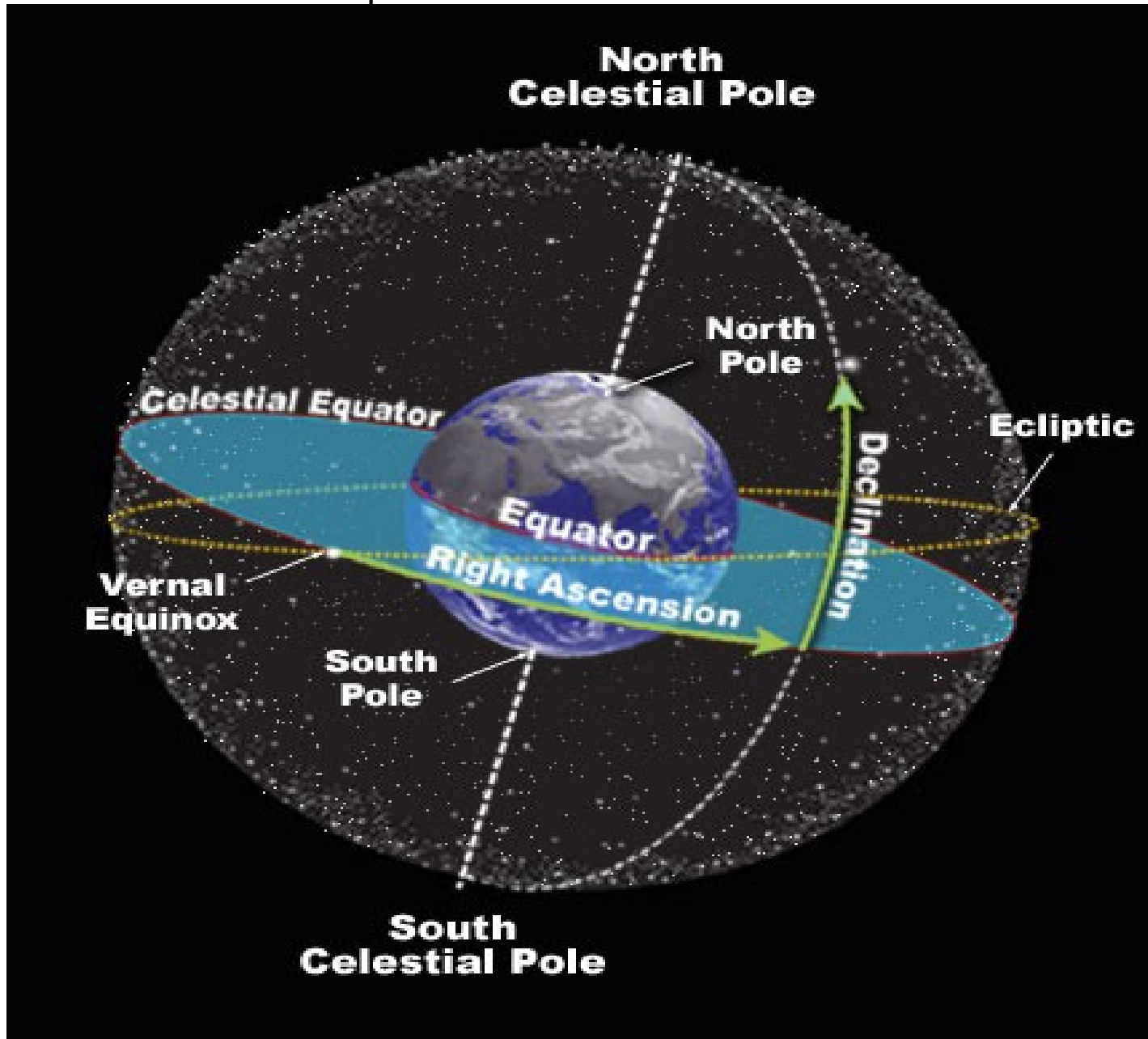
Circumpolar stars stay above equator
See animations motion stars



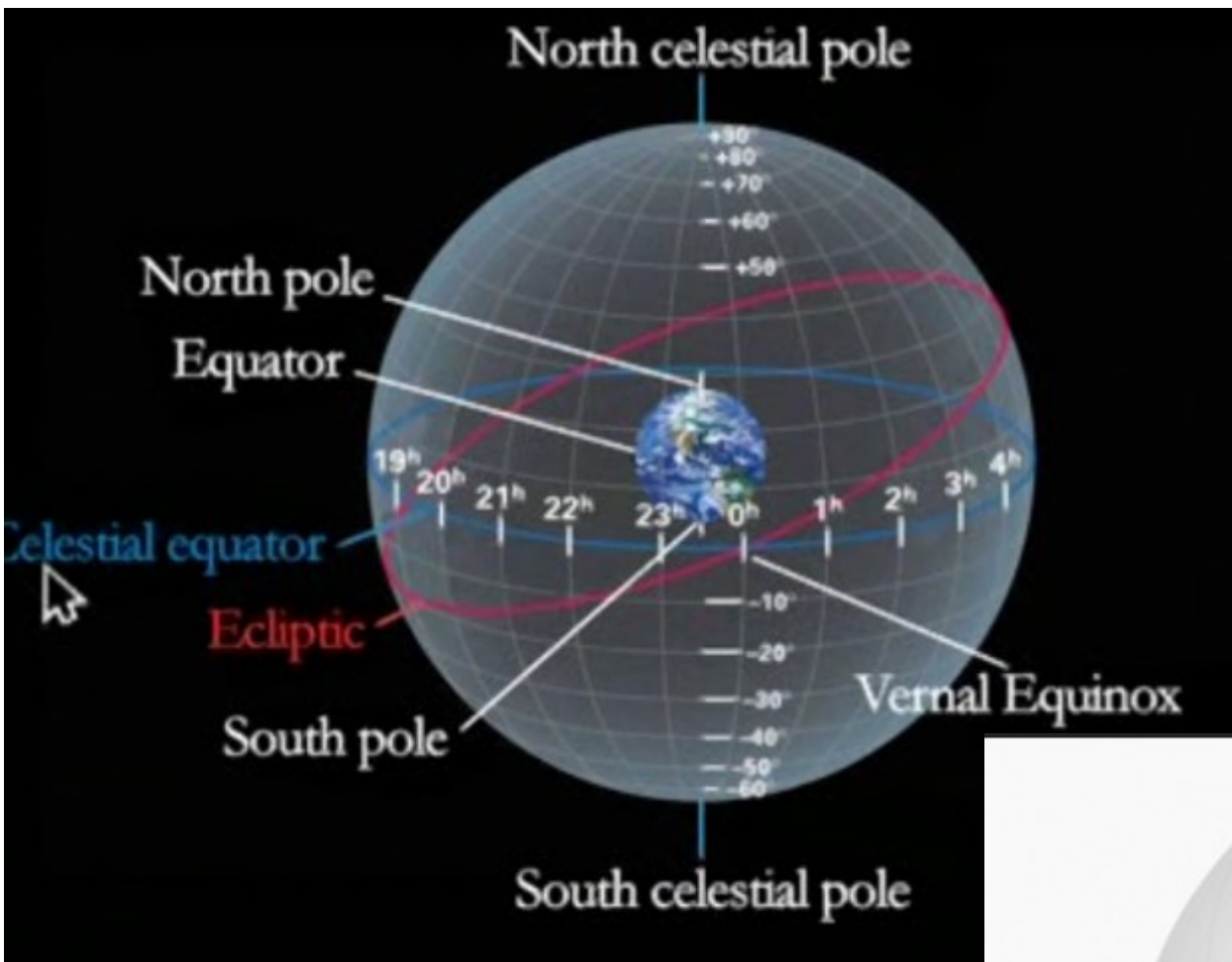
applet



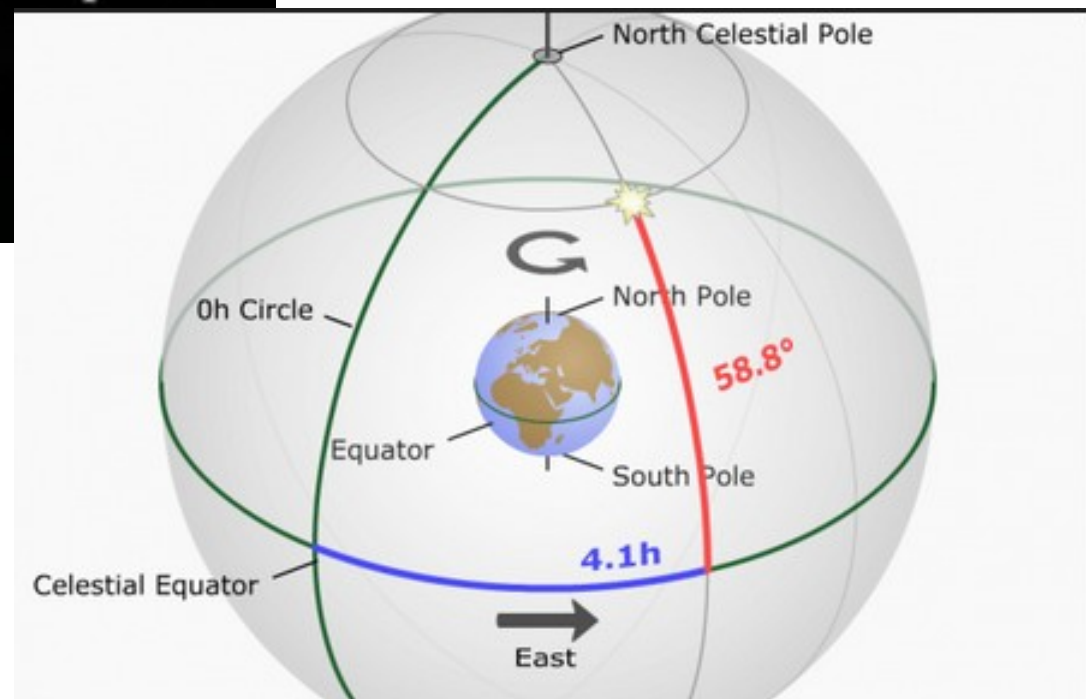
Take your locator. Find the intersections of the equator and the ecliptic lines. These intersections corresponds to which dates ?

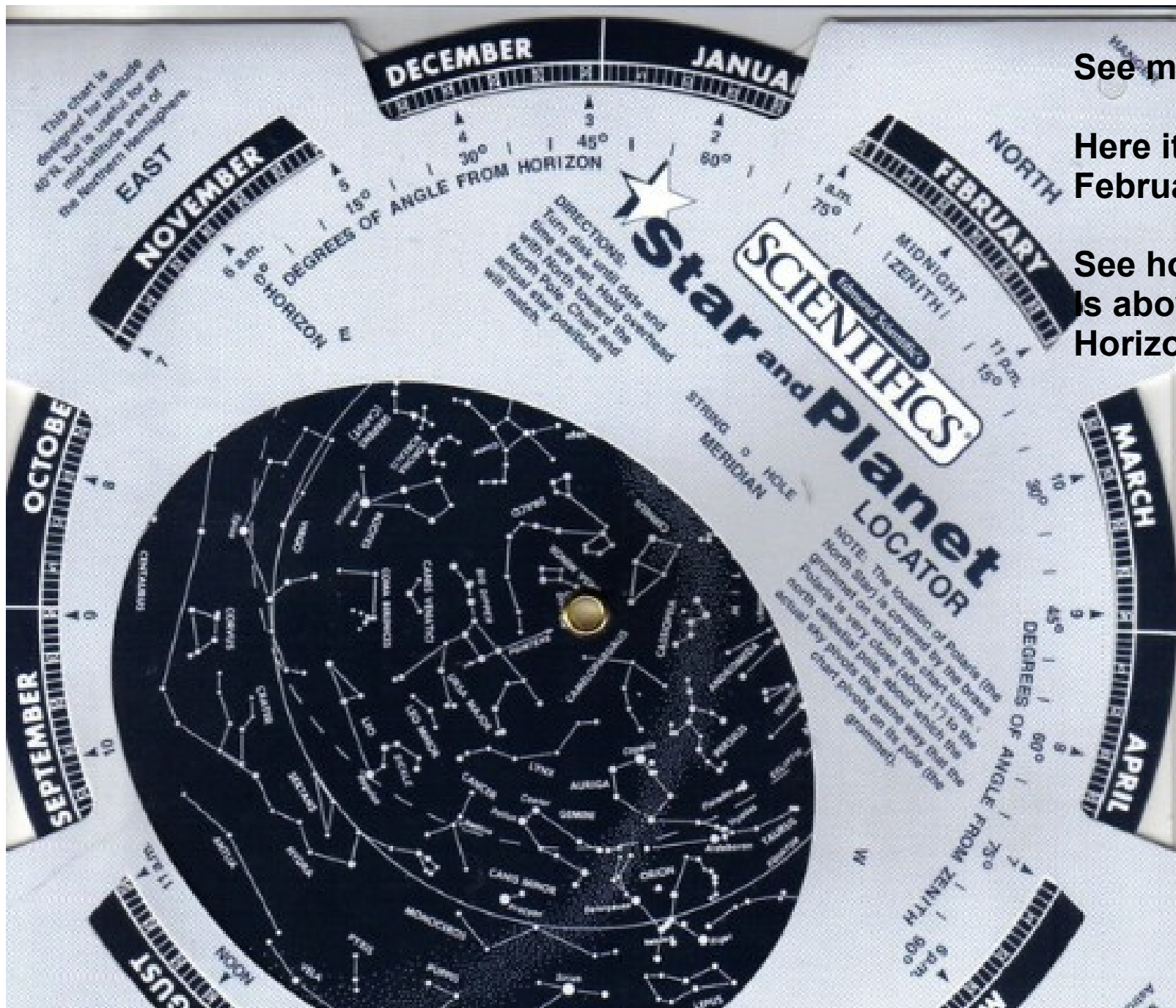


Start from the vernal and go clockwise to find an object. Count in hours. Then go above the equator toward the North star . Count in degrees. Altair 19h49m +8.78



Latitude = declination
 Longitude = right ascension



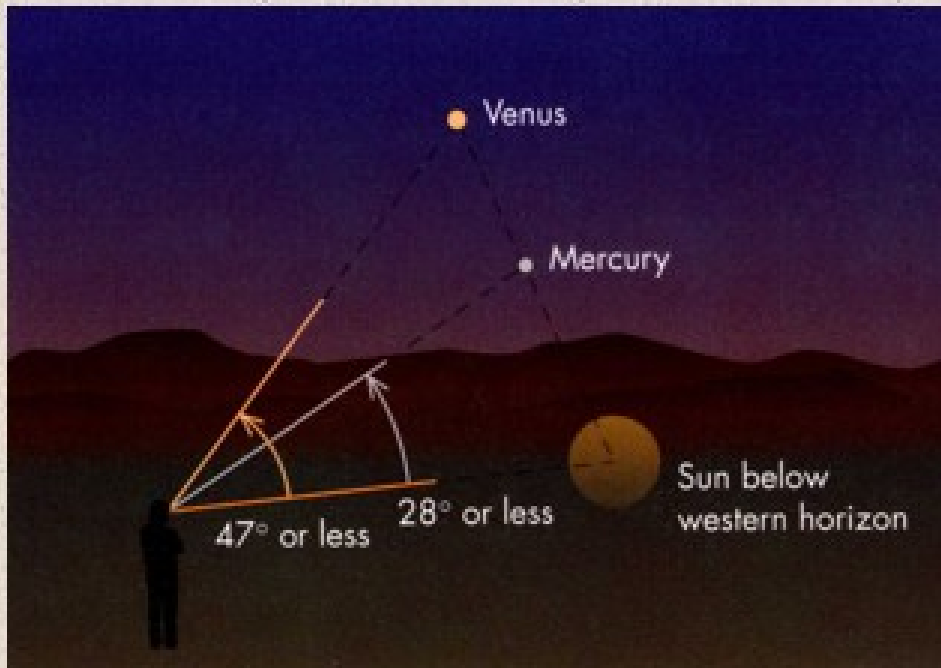


See movie.

Here its 12PM
February

See how polaris
Is above North
Horizon.

Morning and Evening “Stars”



This is because they have orbits inward of ours. That means they can only be seen to a certain maximum angle away from the Sun.

We see Mercury and Venus follow the Sun around in the sky. They may go down after, or come up before it. If they go down after, we see them in the evening.

